

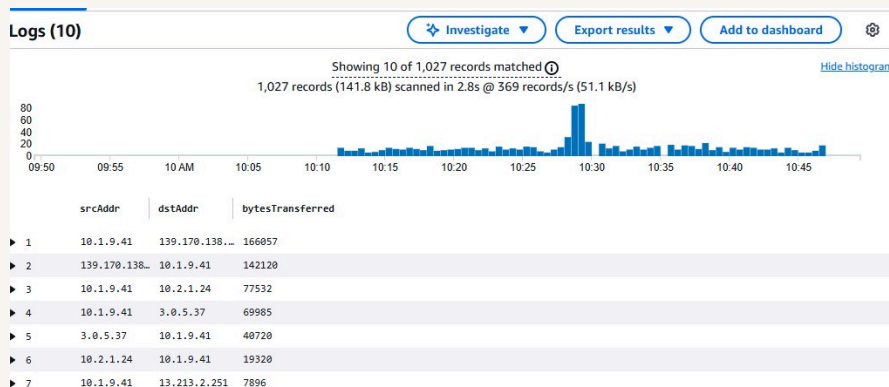


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VPC Monitoring with Flow Logs



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Introducing Today's Project!

What is Amazon VPC?

Amazon Virtual Private Cloud (VPC) allows you to launch AWS resources into a logically isolated virtual network that you define, providing enhanced security, control, and flexibility over your cloud environment.

How I used Amazon VPC in this project

I used two VPC, used VPC peering to see the logs and get insights.

One thing I didn't expect in this project was...

Too much screenshots...

This project took me...

It took me 40 minutes to complete this project.



In the first part of my project...

Step 1 - Set up VPCs

I'll start from creating two VPCs.

Step 2 - Launch EC2 instances

I am going to launch EC2 instance to test the VPC peering.

Step 3 - Set up Logs

In this step I am going set up flow logs.

Step 4 - Set IAM permissions for Logs

in this set up we Give VPC Flow Logs the permission to write logs and send them to CloudWatch. Finish setting up your subnet's flow log.



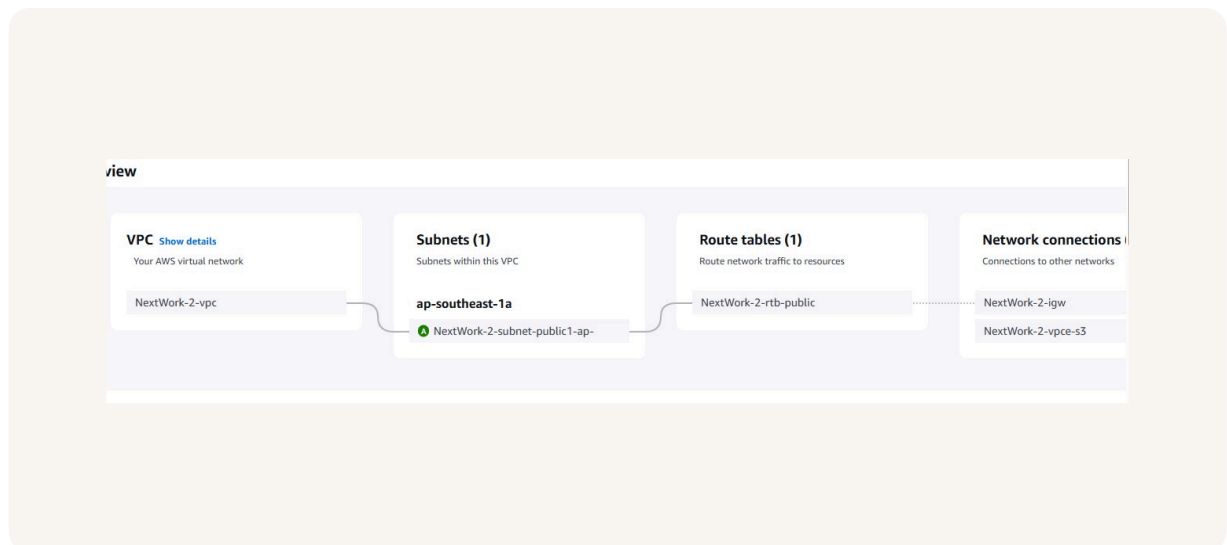
Multi-VPC Architecture

I started my project by launching 2 VPCs, with 1 Public subnet each.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0 and 10.2.0.0. They have to be unique because to avoid network and traffic flow issues.

I also launched EC2 instances in each subnet

My EC2 instances' security groups allow all IP traffic. This is because I want to make it public in order to check the connectivity.

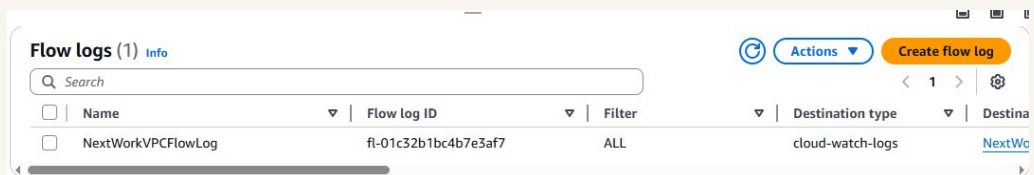




Logs

Logs are simply a record of an AWS service used.

Log groups are like a folder that stores more than one file or log entry.



The screenshot shows the AWS Flow Logs console. At the top, there's a header "Flow logs (1)" with an "Info" link. Below this is a search bar labeled "Search". To the right of the search bar are buttons for "Actions" (with a dropdown arrow) and "Create flow log". Below the search bar is a table with columns: Name, Flow log ID, Filter, Destination type, and Destination. There is one row in the table with the following data: Name: NextWorkVPCFlowLog, Flow log ID: fl-01c32b1bc4b7e3af7, Filter: ALL, Destination type: cloud-watch-logs, and Destination: NextWorkVPCFlowLog. The table has a scrollbar at the bottom.

☐	Name	Flow log ID	Filter	Destination type	Destination
☐	NextWorkVPCFlowLog	fl-01c32b1bc4b7e3af7	ALL	cloud-watch-logs	NextWorkVPCFlowLog



IAM Policy and Roles

I created an IAM policy because it is important to give permissions to flow log via IAM role to record the flow logs.

I also created an IAM role because it is a user that is going to use this policy.

A custom trust policy is specific type of policy! They're different from IAM policies. While IAM policies help you define the actions a user/service can or cannot do, custom trust policies are used to very narrowly define who can use a role

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "Statement1",  
6       "Effect": "Allow",  
7       "Principal": {  
8         "Service": "vpc-flow-logs.amazonaws.com"  
9       },  
10      "Action": "sts:AssumeRole"  
11    }  
12  ]  
13 }
```



In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step we get Instance 1 to send test messages to Instance 2

Step 6 - Set up a peering connection

in this step we are going to set up a connection link between your VPCs for VPC peering.

Step 7 - Analyze flow logs

Review the flow logs recorded about VPC 1's public subnet. Analyse the flow logs to get some tasty insights



Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means the connectivity failed.

```
PING 10.2.1.24 (10.2.1.24) 56(84) bytes of data.  
□
```

I could receive ping replies if I ran the ping test using the other instance's public IP address, which means it is correctly configured, and can go on public internet.

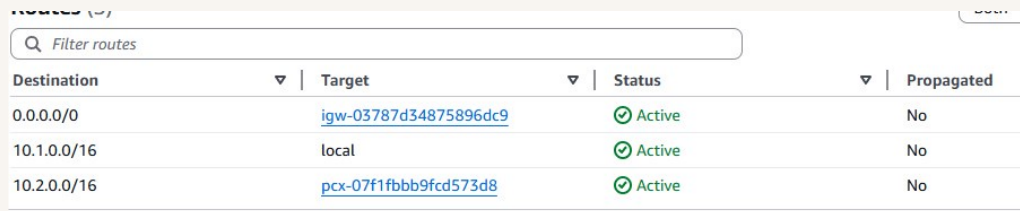


Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because VPC peering connection is missing

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that both the VPCs can communicate with each other.



The screenshot shows the 'Routes' tab of an AWS route table. It contains a search bar 'Filter routes' and a table with the following data:

Destination	Target	Status	Propagated
0.0.0.0/0	igw-03787d34875896dc9	Active	No
10.1.0.0/16	local	Active	No
10.2.0.0/16	pcx-07f1fbbb9fcd573d8	Active	No



Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means correct configuration and both VPCs can communicate with each other.

```
64 bytes from 10.2.1.24: icmp_seq=1 ttl=127 time=0.952 ms
64 bytes from 10.2.1.24: icmp_seq=2 ttl=127 time=0.288 ms
64 bytes from 10.2.1.24: icmp_seq=3 ttl=127 time=1.02 ms
64 bytes from 10.2.1.24: icmp_seq=4 ttl=127 time=0.922 ms
64 bytes from 10.2.1.24: icmp_seq=5 ttl=127 time=1.36 ms
64 bytes from 10.2.1.24: icmp_seq=6 ttl=127 time=0.776 ms
64 bytes from 10.2.1.24: icmp_seq=7 ttl=127 time=0.669 ms
64 bytes from 10.2.1.24: icmp_seq=8 ttl=127 time=1.12 ms
64 bytes from 10.2.1.24: icmp_seq=9 ttl=127 time=0.399 ms
64 bytes from 10.2.1.24: icmp_seq=10 ttl=127 time=0.468 ms
64 bytes from 10.2.1.24: icmp_seq=11 ttl=127 time=1.19 ms
64 bytes from 10.2.1.24: icmp_seq=12 ttl=127 time=0.313 ms
64 bytes from 10.2.1.24: icmp_seq=13 ttl=127 time=0.608 ms
64 bytes from 10.2.1.24: icmp_seq=14 ttl=127 time=0.600 ms
64 bytes from 10.2.1.24: icmp_seq=15 ttl=127 time=0.559 ms
64 bytes from 10.2.1.24: icmp_seq=16 ttl=127 time=1.20 ms
64 bytes from 10.2.1.24: icmp_seq=17 ttl=127 time=0.563 ms
64 bytes from 10.2.1.24: icmp_seq=18 ttl=127 time=0.351 ms
64 bytes from 10.2.1.24: icmp_seq=19 ttl=127 time=1.11 ms
64 bytes from 10.2.1.24: icmp_seq=20 ttl=127 time=0.356 ms
64 bytes from 10.2.1.24: icmp_seq=21 ttl=127 time=0.883 ms
```



Analyzing flow logs

Flow logs tell us about as source and destination IP addresses, source and destination ports, protocol, number of packets and bytes transferred, start and end times of the flow, the action taken (e.g., accept or reject), and a log s

For example, the flow log I've captured tells us when, where, and who is performing the activity like login, logout etc.

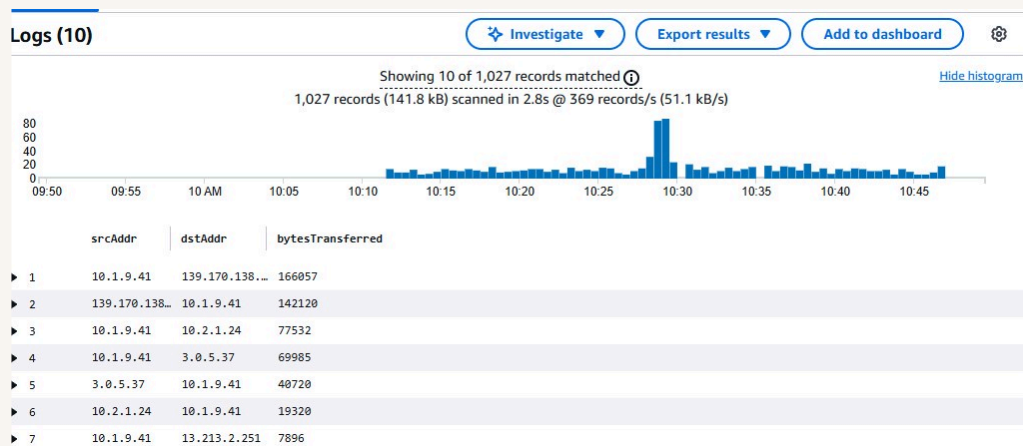
▶	Timestamp	Message
		There are older events to load. Load more.
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 198.55.98.201 10.1.9.41 43949 32160 6 1 40 1748342092 1748342131 REJ...
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 92.63.197.179 10.1.9.41 52758 7817 6 1 40 1748342092 1748342131 REJE...
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 42.112.16.27 10.1.9.41 52348 6218 6 1 44 1748342092 1748342131 REJEC...
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 89.248.165.136 10.1.9.41 43737 2488 6 1 40 1748342092 1748342131 REJ...
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 45.33.89.53 10.1.9.41 34294 6666 6 1 44 1748342092 1748342131 REJECT...
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 8.222.131.41 10.1.9.41 41050 1337 6 1 44 1748342092 1748342131 REJEC...
▶	2025-05-27T10:34:52.000Z	2 120569621470 eni-099d1e8bb685f3493 42.112.16.27 10.1.9.41 52348 6782 6 1 44 1748342092 1748342131 REJEC...



Logs Insights

AWS CloudWatch Logs Insights is a powerful service that allows you to interactively search, analyze, and visualize your log data within Amazon CloudWatch Logs. It helps you efficiently respond to operational issues, identify potential causes, etc.

I ran the query was for 10 logs. This query analyzes top 10 source and destinations IPs where most of the data was tranffered.





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